

Ch 2: Acids, Bases and Salts

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

- Q1. What is a universal indicator? What does it show? How is it different from a litmus indicator?**
- Q2. What is the pH scale? What does a pH of 7, less than 7, and more than 7 indicate?**
- Q3. Why does dry HCl gas not change the colour of dry litmus paper? (PYQ 2022)**
- Q4. What happens when an acid is dissolved in water? What is this process called?**
- Q5. What is a neutralisation reaction? Give one example with a balanced chemical equation. (PYQ 2021)**
- Q6. What is a salt? How is it formed? Give two examples of salts. (PYQ 2020)**
- Q7. Name the gas evolved when zinc reacts with dilute sulphuric acid. Write the balanced chemical equation. (PYQ 2019)**
- Q8. What happens when sodium hydroxide solution is mixed with hydrochloric acid? Name the type of reaction. Write the balanced equation.**
- Q9. Why does an aqueous solution of an acid conduct electricity? (PYQ 2022)**
- Q10. What are strong acids and weak acids? Give one example of each. (PYQ 2023)**
- Q11. What is the chemical name and formula of washing soda? Write one use of washing soda. (PYQ 2020)**
- Q12. What is the chemical name and formula of baking soda? Write one use of baking soda. (PYQ 2019)**

Q13. What is the chemical name and formula of bleaching powder? Write its one use. (PYQ 2018)

Q14. What is the chemical name and formula of Plaster of Paris? Write one use of Plaster of Paris. (PYQ 2019)

Q15. What is water of crystallisation? Give one example of a salt that has water of crystallisation. (PYQ 2022)

Q16. What happens when blue vitriol ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) is heated? Write the chemical equation. (PYQ 2020)

Q17. What happens when Plaster of Paris is mixed with water? Write the chemical equation and state one use.

Q18. Why does tooth decay occur? What role does pH play in it? (PYQ 2019)

Q19. What is the pH of human blood? What happens if the pH changes even slightly? (PYQ 2021)

Q20. Name the acid present in the following: (i) Vinegar (ii) Ant sting (iii) Tomato (iv) Curd (PYQ 2020)

Q21. What is the action of litmus on the following solutions: (i) NaOH (ii) HCl (iii) NaCl (iv) Na_2CO_3 ? (PYQ 2019)

Q22. What gas is evolved when a metal carbonate reacts with an acid? Write the balanced equation for the reaction of Na_2CO_3 with HCl. (PYQ 2018)

Q23. What is the common name and chemical formula of: (i) Common salt (ii) Caustic soda (iii) Caustic potash? (PYQ 2022)

Q24. Why does an aqueous solution of sodium carbonate turn red litmus blue? (PYQ 2023)

Q25. What is olfactory indicator? Give one example. (PYQ 2024)

Q26. What is the difference between the action of acid on metal and on metal carbonate? Write one equation each.

Q27. Why should acids always be added to water and not water to acid? (PYQ 2019)

Q28. Write the balanced chemical equation for the reaction of zinc with sodium hydroxide. Name the gas evolved. (PYQ 2020)

Q29. What is the role of baking soda in making cakes and bread light and fluffy? (PYQ 2022)

Q30. A solution turns red litmus blue. Its pH is likely to be 3, 7, 8, or 11? Choose the correct answer and give a reason. (PYQ 2023)

Q31. What is the relationship between the concentration of H_3O^+ ions and OH^- ions in a neutral solution? (PYQ)

Q32. Why is sodium hydrogen carbonate used in antacids? (PYQ 2019)

SECTION B - 3 MARKS

Q1. Write the balanced chemical equations for the reactions of dilute hydrochloric acid with the following: (PYQ 2019)

Substance	Balanced Chemical Equation	Name of Salt Formed
Iron (Fe)		
Marble chips (CaCO_3)		
Sodium hydroxide (NaOH)		

Q2. Write the balanced chemical equations for the reactions of dilute sulphuric acid with the following: (PYQ 2020)

Substance	Balanced Chemical Equation	Observation
Zinc (Zn)		
Copper oxide (CuO)		
Sodium carbonate (Na_2CO_3)		

Q3. Complete the following table showing reactions of bases with different substances: (PYQ 2018)

Substance Reacting with Base	Balanced Chemical Equation	Products Formed
Acids ($\text{NaOH} + \text{HCl}$)		
Non-metallic oxide ($\text{NaOH} + \text{CO}_2$)		
Metals ($\text{NaOH} + \text{Zn}$)		

Q4. Match the following salts with their chemical name, formula, and one use: (PYQ 2022)

Common Name	Chemical Name	Chemical Formula	One Use
Washing soda			
Baking soda			
Bleaching powder			

Common Name	Chemical Name	Chemical Formula	One Use
Plaster of Paris			

Q5. Write the balanced chemical equation and state the observation for each of the following reactions: (PYQ 2021)

Reaction	Balanced Equation	Observation
Dilute HCl added to zinc granules		
Dilute HCl added to sodium carbonate		
Sodium hydroxide added to copper sulphate solution		

Q6. Fill in the following table about the pH of common substances and their nature: (PYQ 2020)

Substance	pH Value	Nature (Acidic / Basic / Neutral)	Colour on Universal Indicator
Human blood	7.4		
Lemon juice	2-3		
Milk of magnesia	10		
Pure water	7		
Gastric juice (HCl)	1-2		

Q7. Write the balanced chemical equations for the preparation of the following salts: (PYQ 2019)

Salt to be Prepared	Reactants Used	Balanced Chemical Equation
Sodium chloride (NaCl)		
Copper sulphate (CuSO ₄)		
Sodium sulphate (Na ₂ SO ₄)		

Q8. Write the balanced chemical equations for the following and name the type of salt formed in each reaction: (PYQ 2022)

Reaction	Balanced Equation	Salt Formed	Type (Acidic/Basic/Neutral)
HCl + NaOH			
H ₂ SO ₄ + Ca(OH) ₂			
HNO ₃ + KOH			

Q9. What is the importance of pH in everyday life? Explain with any three examples. (PYQ 2019)

Q10. Write the balanced chemical equations for the following reactions. Also identify the type of reaction in each case: (PYQ 2023)

Reaction	Balanced Chemical Equation	Type of Reaction
Dilute HCl + magnesium ribbon		
Sodium hydroxide + CO ₂		
Copper oxide + H ₂ SO ₄		

Q11. Explain what happens when: (i) excess CO₂ is passed through lime water (ii) baking soda is heated (iii) washing soda is exposed to open air. Write balanced equations for each. (PYQ 2020)

Q12. What is the action of the following substances on litmus? Also state whether each solution is acidic, basic, or neutral: (PYQ 2021)

Substance	Action on Red Litmus	Action on Blue Litmus	Nature
HCl solution			
NaOH solution			
NaCl solution			
Na ₂ CO ₃ solution			
Lemon juice			

Q13. What is the role of the following in everyday life? Write the chemical formula and one chemical reaction for each: (PYQ 2019)

Substance	Chemical Formula	Role in Everyday Life	Chemical Reaction
Baking soda			
Washing soda			
Bleaching powder			

Q14. Write balanced chemical equations for the reactions of the following metals with sodium hydroxide solution: (PYQ 2020)

Metal	Balanced Chemical Equation	Name of Gas Evolved	Name of Salt Formed
Aluminium (Al)			
Zinc (Zn)			

Q15. A student is given three solutions A, B, and C. Their pH values are 3, 7, and 11 respectively. Answer the following: (PYQ 2022)

Question	Answer
Which solution is acidic?	
Which solution is basic?	
Which solution is neutral?	
Which solution turns red litmus blue?	
Which solution will react with NaOH?	

Q16. Write the balanced equations for the reactions of dilute HCl with the following metallic oxides and state the colour of the salt formed: (PYQ)

Metallic Oxide	Balanced Equation	Salt Formed	Colour of Salt Solution
CuO			
Fe ₂ O ₃			
ZnO			

Q17. Write the balanced chemical equations for the following double displacement reactions. State whether a precipitate is formed and name it: (PYQ 2019)

Reaction	Balanced Equation	Precipitate Formed
$\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq})$		
$\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{KI}(\text{aq})$		
$\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq})$		

Q18. What is a self-ionisation of water? What is the product of $[\text{H}^+]$ and $[\text{OH}^-]$ concentrations? What does this tell us about the nature of the solution? (PYQ 2020)

Q19. Explain what happens when: (i) copper sulphate crystals are heated (ii) water is added to anhydrous copper sulphate (iii) Plaster of Paris is mixed with water. Write equations. (PYQ 2021)

Q20. Write the balanced equations for the following reactions of acids with metal carbonates and bicarbonates: (PYQ 2019)

Reaction	Balanced Chemical Equation	Gas Evolved
$\text{HCl} + \text{Na}_2\text{CO}_3$		
$\text{HCl} + \text{NaHCO}_3$		
$\text{H}_2\text{SO}_4 + \text{CaCO}_3$		

Q21. Define the following with one example and chemical equation each: (PYQ 2023)

Term	Definition	Example with Equation
Acid		
Base		
Salt		

Q22. Explain with balanced equations what happens when: (i) dilute H_2SO_4 is added to barium chloride solution (ii) excess NaOH is added to ZnSO_4 solution (iii) dilute HNO_3 is added to egg shells. (PYQ 2022)

Q23. Compare the properties of acids and bases: (PYQ 2018)

Property	Acids	Bases
Taste		
Effect on red litmus		
Effect on blue litmus		
pH range		
Example		

Q24. What is the importance of the chlor-alkali process? Write the equation for electrolysis of brine and name all the products and their uses. (PYQ 2020)

Q25. What are indicators? Name three indicators used in the laboratory and state what colour change they show in acidic and basic solutions: (PYQ 2019)

Indicator	Colour in Acidic Solution	Colour in Basic Solution	Colour in Neutral Solution
Litmus			
Methyl orange			
Phenolphthalein			

Q26. What is the difference between a dilute acid and a concentrated acid? What safety precaution must be taken while diluting concentrated H_2SO_4 ? Why? (PYQ 2019)

Q27. Explain the preparation of bleaching powder. Write the chemical equation. Name its active ingredient. What happens when bleaching powder reacts with dilute H_2SO_4 ? (PYQ 2019)

Q28. Why do metallic oxides react with acids while non-metallic oxides react with bases? Write one balanced equation for each type of reaction. (PYQ 2022)

Q29. Write the balanced chemical equations for the following: (PYQ 2020)

Reaction	Balanced Chemical Equation	Type of Reaction
Dilute H_2SO_4 + zinc		
Dilute HCl + iron		

Reaction	Balanced Chemical Equation	Type of Reaction
Dilute HNO_3 + aluminium		

Q30. A student tests four solutions with red and blue litmus paper and a pH meter. Fill in the table and classify each solution: (PYQ 2023)

Solution	pH Value	Effect on Red Litmus	Effect on Blue Litmus	Classification
HCl	1			
NaOH	13			
NaCl	7			
CH_3COOH	4			

SECTION C - 5 MARKS

Q1. Write the balanced chemical equations for the reactions of dilute hydrochloric acid with the following. Also name the type of reaction in each case: (PYQ 2019)

Substance	Balanced Equation	Type of Reaction	Observation
Magnesium ribbon			
Aluminium metal			
Iron filings			
Copper oxide (CuO)			
Sodium carbonate (Na ₂ CO ₃)			

Q2. Write the balanced equations for the following reactions of bases with different substances. State one observation for each: (PYQ 2020)

Substance Reacting with NaOH	Balanced Equation	Observation
Dilute HCl (acid)		
CO ₂ (non-metallic oxide)		
Zinc (metal)		
Aluminium (metal)		
Copper sulphate solution (CuSO ₄)		

Q3. Complete the following table about important salts: their chemical names, formulas, preparation equations, and uses: (PYQ 2022)

Common Name	Chemical Name	Formula	Preparation Equation	One Important Use
Common salt				
Washing soda				
Baking soda				
Bleaching powder				
Plaster of Paris				

Q4. Write balanced chemical equations for the following and identify the type of salt (acidic, basic, or neutral) formed in each: (PYQ 2021)

Reaction	Balanced Equation	Salt Formed	Type of Salt	pH of Salt Solution
HCl + NaOH				
H ₂ SO ₄ + 2NaOH				
HNO ₃ + KOH				
HCl + NH ₄ OH				
CH ₃ COOH + NaOH				

Q5. Explain the importance of pH in everyday life with five different examples. Also state what happens when the pH of rain water falls below 5.6: (PYQ 2023)

Situation	Normal pH	Effect of Abnormal pH	Remedy
Human blood			
Tooth decay			
Soil for plants			
Bee sting / Ant sting			
Acid rain			

Q6. Write the balanced chemical equations for the following reactions. Classify each reaction type and state one observation: (PYQ 2024)

Reaction	Balanced Equation	Type	Observation
Dilute HCl added to zinc granules			
NaOH added to FeSO ₄ solution			
Baking soda heated strongly			
Excess CO ₂ passed through lime water			
Dilute H ₂ SO ₄ added to BaCl ₂ solution			

Q7. A student performs the following experiments in the laboratory. State the observation, write the balanced equation, and name the type of reaction: (PYQ 2020)

Experiment	Observation	Balanced Equation	Type of Reaction
Dilute HCl dropped on marble chips (CaCO_3)			
NaOH solution + HCl solution mixed			
Zinc added to NaOH solution			
Copper oxide added to H_2SO_4			
Dilute HCl added to Na_2CO_3 solution			

Q8. Explain the chlor-alkali process. Write the balanced equation for the electrolysis of brine. Name all three products and write one chemical equation and one use for each product: (PYQ 2020)

Product	State / Nature	Chemical Equation Showing Its Use	Practical Use
Chlorine gas (Cl_2)			
Hydrogen gas (H_2)			
Sodium hydroxide (NaOH)			

Q9. Answer the following questions about acids and bases: (PYQ 2022)

Question	Answer
Why does HCl dissociate in water but not in benzene?	
What is the common property of all acids?	
Why does NaOH solution conduct electricity?	
What is the pH of a solution with $[\text{H}^+] = [\text{OH}^-]$?	
Why does a solution of CO_2 in water turn blue litmus red?	

Q10. Write balanced chemical equations for the preparation of the following salts. Also state one use of each: (PYQ 2019)

Salt	Method of Preparation	Balanced Equation	One Use
Sodium chloride (NaCl)	Acid + base		
Copper sulphate (CuSO ₄)	Acid + metallic oxide		
Zinc sulphate (ZnSO ₄)	Acid + metal		
Calcium chloride (CaCl ₂)	Acid + metal carbonate		
Ferrous sulphate (FeSO ₄)	Metal + acid		

Q11. Write the balanced chemical equations and state the observation for the following reactions. Identify the type of each reaction: (PYQ 2023)

Reaction	Balanced Equation	Observation	Type
Mg ribbon + dilute HCl			
NaOH + AlCl ₃ solution			
Ca(OH) ₂ + Cl ₂			
Na ₂ CO ₃ + Ca(OH) ₂ solution			
NaHCO ₃ heated			

Q12. Explain the following with balanced chemical equations: (PYQ 2024 case-based)

Situation	Explanation	Balanced Chemical Equation
Why does baking soda make bread rise?		
Why is washing soda used to remove hardness of water?		
Why is bleaching powder used to disinfect water?		
Why is Plaster of Paris used in making casts?		
Why is antacid used to treat acidity in stomach?		

Q13. A student tests solutions A (pH 2), B (pH 7), C (pH 10), and D (pH 14) with litmus and universal indicator. Complete the table and answer the questions: (PYQ 2022)

Solution	pH	Nature	Colour on Universal Indicator	Effect on Red Litmus	Example Substance
A	2				
B	7				
C	10				
D	14				

Q14. Write balanced chemical equations for the following reactions. State whether each reaction is exothermic or endothermic: (PYQ 2020)

Reaction	Balanced Equation	Exothermic or Endothermic	Reason
H ₂ SO ₄ dissolved in water			
NaOH + HCl neutralisation			
Thermal decomposition of NaHCO ₃			
Ca(OH) ₂ + CO ₂ → CaCO ₃ + H ₂ O			
Dilute HCl + Mg ribbon			

Q15. Answer the following questions about water of crystallisation and important salts: (PYQ 2024 case-based)

Question	Answer
What is water of crystallisation? Give one example.	
What happens when CuSO ₄ .5H ₂ O is heated? Write the equation.	
What is the chemical name and formula of Plaster of Paris?	
What happens when Plaster of Paris is mixed with water?	
Why does anhydrous CuSO ₄ turn blue when water is added?	